

Topic Selection:

Research Proposal Title: Healthcare Analysis

About Medicare:

In the United States, Medicare is a single-payer, national social insurance program administered by the U.S. federal government since 1966. United States Medicare is funded by a payroll tax, premiums and surtaxes from beneficiaries, and general revenue. It provides health insurance for Americans aged 65 and older who have worked and paid into the system through the payroll tax. It also provides health insurance to younger people with some disability status as determined by the Social Security Administration, as well as people with end stage renal disease and amyotrophic lateral sclerosis.

In 2015, Medicare provided health insurance for over 55 million (46 million people age 65 and older and 9 million younger people). On average, Medicare covers about half of the health care charges for those enrolled. The enrollees must then cover their remaining costs either with supplemental insurance, separate insurance, or out-of-pocket. Out-of-pocket costs can vary depending on the amount of health care a Medicare enrollee needs.

About CMS:

The Centers for Medicare & Medicaid Services (CMS), previously known as the Health Care Financing Administration (HCFA), is a federal agency within the United States Department of Health and Human Services (HHS) that administers the Medicare program and works in partnership with state governments to administer Medicaid, the Children's Health Insurance Program (CHIP), and health insurance portability standards. In addition to these programs, CMS has other responsibilities, including the administrative simplification standards from the Health Insurance Portability and Accountability Act of 1996 (HIPAA), quality standards in long-term care facilities (more commonly referred to as nursing homes) through its survey and certification process, clinical laboratory quality standards under the Clinical Laboratory Improvement Amendments, and oversight of HealthCare.gov.

Research Idea:

Why Did You Choose This Topic?

The choice of "Healthcare Analysis " as a research topic was motivated by several compelling factors. Firstly, healthcare is a critical and rapidly evolving field with immense potential for improving patient care, reducing costs, and enhancing operational efficiency. Secondly, the advent of big data and cloud computing technologies like Google Big Query has opened up unprecedented opportunities for in-depth analysis of healthcare data. This convergence of healthcare and data analytics presents a unique and promising avenue for research.

Moreover, healthcare generates vast amounts of data, including electronic health records, medical images, patient demographics, and more. These data sources hold valuable insights that, when properly analyzed, can lead to better diagnoses, treatment plans, resource allocation, and overall healthcare

outcomes. The ability to harness this data using advanced analytics tools like Big Query aligns with the broader trend of data-driven decision-making in the healthcare industry.

How Is This Research Going to Help or Lead to Business?

The research on "Healthcare Analysis " is poised to have several positive impacts on both healthcare providers and businesses operating within the healthcare ecosystem:

1. **Improved Patient Care:** Deep analysis of healthcare data can lead to more accurate diagnoses, personalized treatment plans, and early detection of diseases. This can ultimately result in improved patient outcomes, reduced hospital stays, and lower healthcare costs.
2. **Operational Efficiency:** Healthcare providers can optimize their operations by using data analytics to streamline processes, allocate resources more effectively, and reduce administrative overhead. This can result in cost savings and increased profitability.
3. **Research and Development:** Pharmaceutical companies and research institutions can leverage deep healthcare analysis to accelerate drug discovery, clinical trials, and the development of innovative treatments. This can lead to the creation of new revenue streams and market opportunities.
4. **Health Insurance:** Insurance companies can use deep analysis to improve risk assessment, pricing strategies, and fraud detection. This can lead to more competitive insurance products and better customer experiences.

Data Exploration:

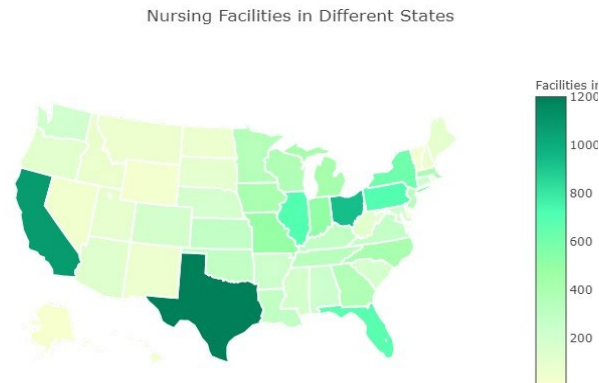
This public dataset was created by the Centers for Medicare & Medicaid Services. The data summarizes the utilization and payments for procedures, services, and prescription drugs provided to Medicare beneficiaries by specific inpatient and outpatient hospitals, physicians, and other suppliers. The dataset includes the following data: Common inpatient and outpatient services, All physician and other supplier procedures and services, All Part D prescriptions. Providers determine what they will charge for items, services, and procedures provided to patients and these charges are the amount that providers bill for an item, service, or procedure.

Points to explore:

- Which State has the highest (and lowest) number of Nursing Facilities
- Average HCC for different states
- States and their average total number stays (of different facilities) and average length of stays (in different facilities)
- Which State has highest claims and Drug Costs
- Which Drugs have been most prescribed in USA?
- Which Drugs are most prescribed in each state?
- Most Common Provider Types in US

1. Which State has the highest (and lowest) number of Nursing Facilities

Skilled nursing facilities (SNFs) and nursing facilities (NFs) are provided by CMS under Medicare. Different States of US have different number of nursing facilities. Number of nursing facilities in a state might be an indicator of how good is the quality of care in different states. Let's obtain the number of nursing facilities using 2014 dataset.



2. Average HCC for different states

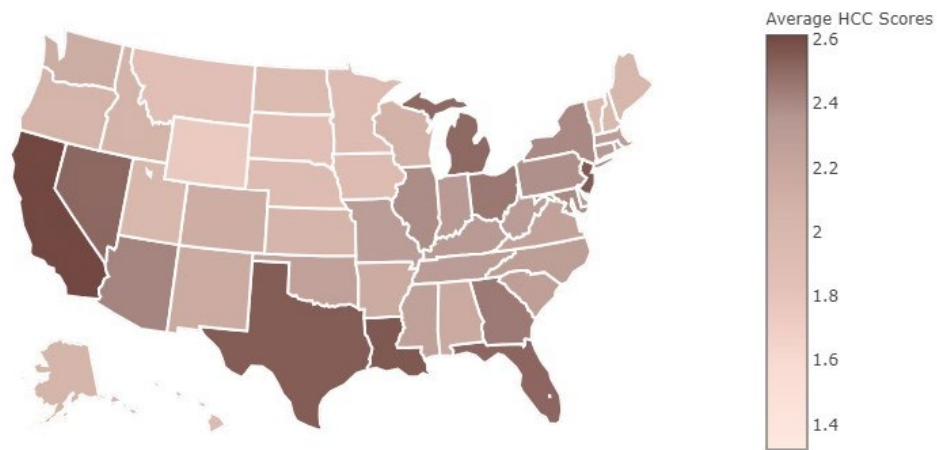
What are HCCs, How and by whom are HCCs used?

HCCs, or Hierarchical Condition Categories, are sets of medical codes that are linked to specific clinical diagnoses. Since 2004, HCCs have been used by the Centers for Medicare and Medicaid Services (CMS) as part of a risk-adjustment model that identifies individuals with serious acute or chronic conditions. This allows Medicare to project the expected risk and future annual cost of care. Each HCC represents diagnoses with similar clinical complexity and expected annual care costs.

HCCs are used to calculate payments to healthcare organizations for patients who are insured by Medicare Advantage (MA) plans, Accountable Care Organizations (ACOs), some Affordable Care Act (ACA) plans and many more. Clinicians add HCCs to a patient's medical record along with supporting documentation as required by CMS.

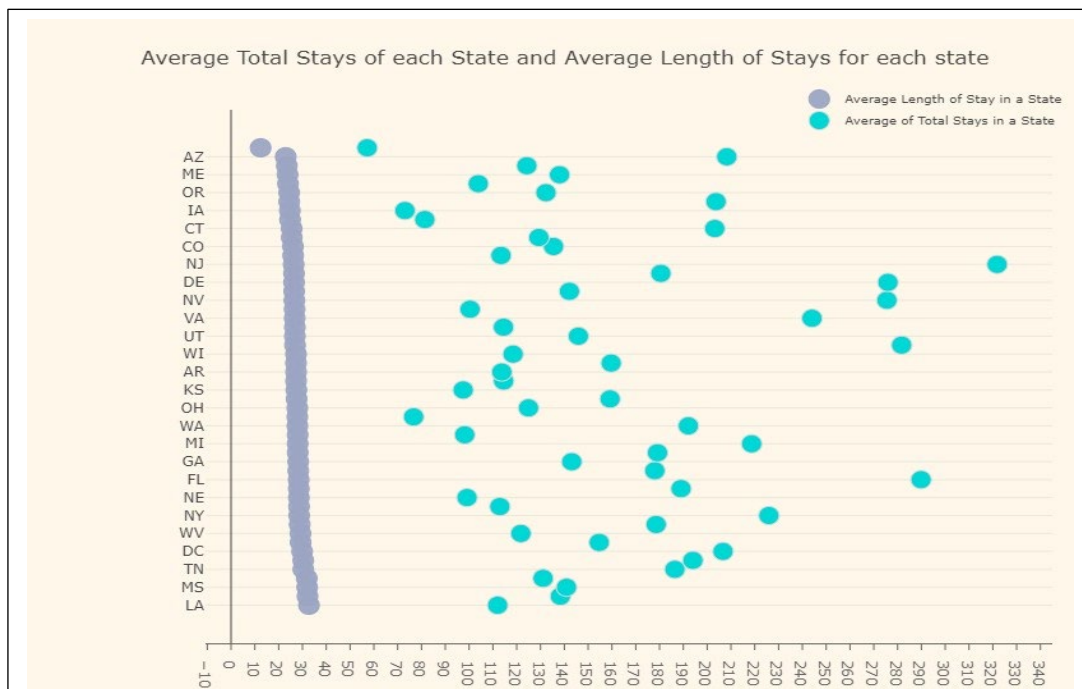
According to below figure, States having highest Average HCC scores are California (2.6), Texas(2.5), Florida(2.5), Minnesota (2.5), Louisiana(2.5), and New Jersey (2.5). States having lowest Average HCC scores are Montana (1.8), South Dakota (1.8), Wyoming (1.7), Puerto Rico (1.3)

Average HCC Scores for different states

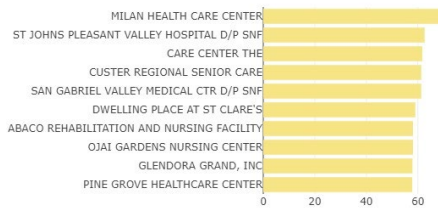


3. States and their average total number stays (of different facilities) and average length of stays (in different facilities):

- Average Length of Stay among the patients in all the states is very close to 30 days. (They also want to avoid 30-day readmission penalty)
- Arizona has the significantly lower average length of stay equal to 22 which is quite less as compared to other states
- Louisiana, Texas, Illinois, Mississippi are the states having greater than 30-day average length of stay, which indicates that patients are admitted there for longer number of days than average
- Puerto Rico, a territory of US has the lowest average length of stay among all the states and territories equal to 12.
- New Jersey's nursing facilities has on-an-average very large number of total stays (equal to 321)
- Iowa has the minimum among the states equal to 73 total stays among its nursing facilities



Nursing Facilities having Maximum Average Length of Stay Days



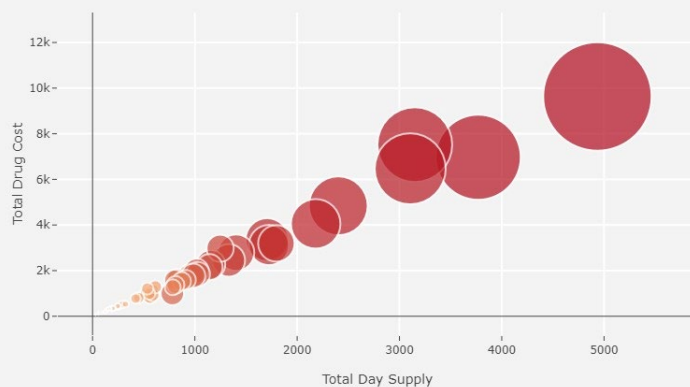
Facilities having highest average length of stay days

4. Which State has highest claims and Drug Costs

California is the state having highest total day supply, highest total drug cost and highest claim count. Other states having more claim counts include Florida, Texas, and New York.

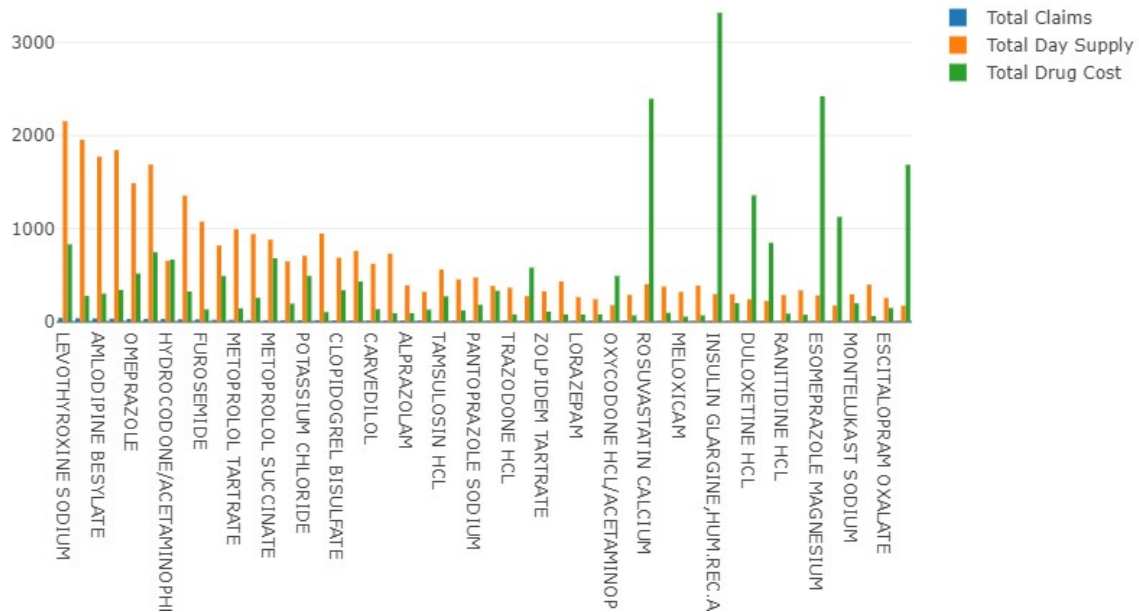
State	Total claim count in millions	Total Drug cost in millions
CA	116	9634
FL	91	6970
NY	80	7522
TX	76	6463
PA	63	4842
OH	53	4051
NC	46	3358
IL	43	3121
MI	39	3194
GA	39	2788

Total claim count VS Total Drug Cost VS Total Day Supply for every state



5. Which Drugs have been most prescribed in US

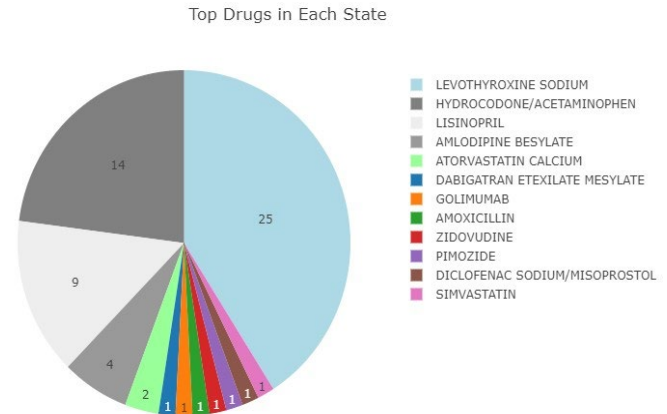
- Levothyroxine Sodium and LISINOPRIL are the drugs which are associated with the highest number of claims as well as highest number of supplies.
- Insulin Glargine, Hum.Rec.An is the drug which is associated with the highest Total Drug Cost
- Drugs having more Total Supply than Drug Costs are AMLODIPINE BESYLATE and SIMVASTATIN
- Drugs having more Total Cost than Supply are Rosuvastatin Calcium, and Esomeprazole magnesium



6. Which Drugs are most prescribed in each state?

In various U.S. states, the most commonly prescribed drugs and their maximum claim counts are as below and these drugs are widely prescribed for various medical conditions across the respective states.

State	Drug Name	Max Claim Count
AA	AMOXICILLIN	225
AE	GOLIMUMAB	666
AK	LISINOPRIL	47684
AL	HYDROCODONE/ACETAMINOPHEN	1214487
AP	PIMOZIDE	208
AR	HYDROCODONE/ACETAMINOPHEN	587718
AS	DICLOFENAC SODIUM/MISOPROSTOL	46
AZ	LEVOTHYROXINE SODIUM	866249
CA	LEVOTHYROXINE SODIUM	3845087
CO	LEVOTHYROXINE SODIUM	644486
CT	ATORVASTATIN CALCIUM	462238
DC	AMLODIPINE BESYLATE	69460
DE	LEVOTHYROXINE SODIUM	100767
FL	LEVOTHYROXINE SODIUM	2982612
GA	AMLODIPINE BESYLATE	1282275
GU	DABIGATRAN ETEXILATE MESYLATE	5271
HI	ATORVASTATIN CALCIUM	158073
IA	LEVOTHYROXINE SODIUM	573390
ID	LEVOTHYROXINE SODIUM	221445
IL	LEVOTHYROXINE SODIUM	1457579
IN	HYDROCODONE/ACETAMINOPHEN	1037637



Knowledge Transfer:

The knowledge generated through the research on "Healthcare Analysis Big" holds the potential to be a catalyst for transformative change within the healthcare industry and various related businesses. Here's how the knowledge derived from this research can be effectively transferred:

Professional Training and Development: Organize training sessions, workshops, and webinars for healthcare professionals, data analysts, and IT specialists. These sessions can teach them how to leverage Google Big Query and advanced data analytics techniques for healthcare-related tasks. By enhancing their skills and knowledge, you empower these professionals to make better use of data in their respective roles.

Educational Resources: Develop educational materials such as online courses, tutorials, and whitepapers that are accessible to a wide audience. These resources can serve as self-paced learning

tools for individuals interested in deep healthcare analysis. They can also be incorporated into academic curricula to train the next generation of healthcare data analysts.